

MAT 528 HW02

Aditya Pahuja

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Problem 1 (Munkres 17.6b). Let A and B denote subsets of a space X . Prove that $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Solution. First, $\overline{A \cup B}$ is a closed set containing A and B , so it contains both $\overline{A}$ and $\overline{B}$, hence $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$. Second, $\overline{A} \cup \overline{B}$ is a closed set containing both A and B , so it contains $A \cup B$ and thus contains its closure, i.e. $\overline{A \cup B} \supseteq \overline{A} \cup \overline{B}$.

We have shown that $\overline{A \cup B}$ and $\overline{A} \cup \overline{B}$ contain each other, so the two sets are equal. $\square$

Problem 2 (Munkres 17.11). Show that the product of two Hausdorff spaces is Hausdorff.

Solution. Let X and Y be Hausdorff spaces and let (x_1, y_1) and (x_2, y_2) be two distinct points in the product space $X \times Y$. Then, at least one of $x_1 \neq x_2$ and $y_1 \neq y_2$ holds.

If the former holds, then since X is Hausdorff we can pick disjoint open sets $U_1 \ni x_1$ and $U_2 \ni x_2$ in X ; then $U_1 \times Y$ and $U_2 \times Y$ are two disjoint open sets in $X \times Y$ (because they are products of open sets) that respectively contain (x_1, y_1) and (x_2, y_2) .

On the other hand, if the latter holds, then since Y is Hausdorff we can pick disjoint open sets $V_1 \ni y_1$ and $V_2 \ni y_2$ in Y ; then $X \times V_1$ and $X \times V_2$ are two disjoint open sets in $X \times Y$ (because they are products of open sets) that respectively contain (x_1, y_1) and (x_2, y_2) .

Therefore, regardless of which two points (x_1, y_1) and (x_2, y_2) in $X \times Y$ are chosen, we can find disjoint open sets in $X \times Y$ separating the two points. $\square$

Problem 3 (Munkres 17.19a). If $A \subseteq X$, we define the **boundary** of A by the equation

$$\partial A = \overline{A} \cap \overline{(X \setminus A)}.$$

Show that $\text{Int}(A)$ and ∂A are disjoint, and $\overline{A} = \text{Int}(A) \cup \partial A$.

Solution. We claim that $\text{Int}(A)$ is the complement of $\overline{(X \setminus A)}$.

If $x \in \text{Int}(A)$, then there is an open subset $U \subseteq X$ such that $x \in U \subseteq A$. This means that U does not intersect $X \setminus A$, so x cannot be in $\overline{(X \setminus A)}$, i.e. $\text{Int}(A)$ is a subset of the complement of $\overline{(X \setminus A)}$.

Conversely, if x is in the complement of $\overline{(X \setminus A)}$, then $x \in A$ and there exists an open set U containing x which does not intersect $X \setminus A$, so $U \subseteq A$; the existence of such a U is what it means for x to be in $\text{Int}(A)$, hence the complement of $\overline{(X \setminus A)}$ is a subset of the interior of A .

We have shown inclusion in both directions, so the claim is proven.

Now, $\text{Int}(A)$ and ∂A must be disjoint because the latter is a subset of the complement $X \setminus \text{Int}(A) = \overline{(X \setminus A)}$, and furthermore

$$\overline{A} = (\overline{A} \cap \overline{(X \setminus A)}) \cup (\overline{A} \cap (X \setminus \overline{(X \setminus A)})) = \partial A \cup (\overline{A} \cap \text{Int}(A)) = \partial A \cup \text{Int}(A)$$

where the last equality is a result of $\text{Int}(A) \subseteq A \subseteq \overline{A}$. $\square$

Problem 4 (Munkres 17.20e). Find the boundary and interior of the set

$$E = \{(x, y) : 0 < x^2 - y^2 \leq 1\} \subseteq \mathbb{R}^2.$$

Solution. Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) = x^2 - y^2$, which is continuous. Then, I claim that $F = f^{-1}([0, 1])$ (which is a closed set containing E by continuity) is the closure of E . The only thing to check is that the points in $f^{-1}(0)$ are all limit points of E , since the rest of the points in F are in E already. If $x^2 - y^2 = 0$ and U is an open set containing (x, y) , then U contains some open ball $B_\varepsilon((x, y))$ centered at (x, y) with $0 < \varepsilon < 1$. Let $r = \min(\varepsilon/(1000x), \varepsilon/1000)$. Then, if x is positive,

$$0 < (x + r)^2 - y^2 = (x + r)^2 - x^2 = 2rx + r^2 \leq \frac{\varepsilon}{500} + \frac{\varepsilon^2}{1000^2} < \varepsilon < 1,$$

and if x is negative we instead bound $(x - r)^2 - y^2$ in the same way, so this ball centered at (x, y) intersects E at a point besides (x, y) (either $(x - r, y)$ or $(x + r, y)$) making (x, y) a limit point of E .

By similar reasoning we can conclude that the points in $f^{-1}(1)$ are also limit points of $\mathbb{R}^2 \setminus E$, meaning any open ball around one of these points will contain a point outside E . Hence $f^{-1}((0, 1))$ is the largest open set contained in E , so it is the interior of E ; thus the boundary of E is $f^{-1}([0, 1]) \setminus f^{-1}((0, 1)) = f^{-1}(\{0, 1\})$. $\square$

Problem 5 (Munkres 18.2). Suppose $f: X \rightarrow Y$ is continuous. If x is a limit point of the subset A of X , is it necessarily true that $f(x)$ is a limit point of $f(A)$?

Solution. No, this statement may not hold; for example, consider $f: \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 17$ for all x . This is continuous because the pre-image of an open set in $\mathbb{R}$ is either empty (if the set doesn't contain 17) or all of $\mathbb{R}$ (if the set does contain 17). Every point of $\mathbb{R}$ is a limit point of $\mathbb{R}$ essentially trivially, but no point of $\mathbb{R}$ is a limit point of $f(\mathbb{R}) = \{17\}$: since $\{17\}$ is closed, the only possible limit point is 17, and no ball centered at 17 intersects $\{17\}$ at a point other than 17. $\square$

Problem 6 (Munkres 18.5). Show that the subspace (a, b) of $\mathbb{R}$ is homeomorphic with $(0, 1)$ and the subspace $[a, b]$ of $\mathbb{R}$ is homeomorphic with $[0, 1]$.

Solution. Take $f: (a, b) \rightarrow (0, 1)$ defined by $f(x) = \frac{x-a}{b-a}$. This is clearly continuous, and we can compute that its inverse is $f^{-1}(y) = (b-a)y + a$, which is also continuous, so f is a homeomorphism between (a, b) and $(0, 1)$.

We can define $g: [a, b] \rightarrow [0, 1]$ in the same way as $g(x) = \frac{x-a}{b-a}$; the inverse is again the continuous map $g^{-1}(y) = (b-a)y + a$, so g is also a homeomorphism between $[a, b]$ and $[0, 1]$. $\square$

Problem 7 (Munkres 18.12). Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by the equation

$$F(x, y) = \begin{cases} xy/(x^2 + y^2) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0). \end{cases}$$

- (a) Show that F is continuous in each variable separately.
- (b) Compute the function $g: \mathbb{R} \rightarrow \mathbb{R}$ defined by $g(x) = F(x, x)$.
- (c) Show that F is not continuous.

Solution. (a) The function is symmetric in x and y , so we only need to prove this for x . If we fix $y = y_0 \neq 0$, then $f_{y_0}(x) = F(x, y_0) = \frac{xy_0}{x^2 + y_0^2}$ is obviously continuous on $\mathbb{R}$ because its denominator never vanishes. If $y_0 = 0$ on the other hand, then $f_{y_0}(x) = 0$ for all x , which is trivially continuous.

(b) $g(x) = F(x, x) = \frac{x^2}{x^2 + x^2} = \frac{1}{2}$.

(c) The set $f^{-1}(\{\frac{1}{2}\})$ is the pre-image of a closed set, so it should be closed. However, $f(x, y) = \frac{1}{2}$ if and only if $(x, y) \neq 0$ and $x^2 + y^2 = 2xy \iff (x - y)^2 = 0 \iff x = y$, which means that the pre-image is the set $\{(x, x) : x \in \mathbb{R}, x \neq 0\}$, which isn't closed because it has $(0, 0)$ as a limit point (since $B_\varepsilon((0, 0))$ contains $(\varepsilon/2, \varepsilon/2)$ for all $\varepsilon > 0$) but it doesn't contain 0. $\square$